



ECO101: INTRODUCTION TO MICROECONOMICS

ASSIGNMENT 01

Total marks 15

(The Assignment should be done in 3 groups of 3 students, check google classroom for more details)

Question 1

1. Suppose that China and Bangladesh have the outputs per worker in producing t-shirt and Bicycle in a day, shown in the table below:

Country	T-Shirt	Bicycle
China	500	1000
Bangladesh	4000	200

- a. **(1 Marks)** What is China's opportunity cost of producing 500 t-shirts?
- b. **(2 Marks)** Which country should specialize in making Bicycle? Explain your answer using the gains from trade theory.
- c. **(2 Marks)** Which country has an absolute advantage in bicycle and T-shirt production?



Question 2

The demand and supply equation for the market of calculator is given below

$$2P = 100 - Q_d \text{ and, } 3P = Q_s - 20$$

- a. What is the price elasticity of demand if prices increase by 10%? Comment on the price elasticity of the product. (3 marks)
- b. Based on your answer of (a) would you recommend the company to increase their prices? Justify your answer. (2 marks)

Question 3

For each of the following scenarios, draw a demand-supply diagram to illustrate and explain the effects of the given shocks on the equilibrium price and quantity in the specified competitive market. Assume that each market is initially at equilibrium:

- a. **(2.5 Marks)** Initially, the market for laptops was at equilibrium, with the demand for and supply of laptops balanced. However, due to a sudden increase in remote work and online learning, the consumption of laptops surged. Additionally, technological advancements have led to more efficient manufacturing processes, resulting in an increase in the production of laptops. If the consumption of laptops increased more than the impact of technological advancement, describe the impact of these changes in both demand and supply on the market for laptops.
- b. **(2.5 Marks)** A major cocoa-producing country has experienced adverse weather conditions, leading to a decrease in cocoa bean production. How do you expect the equilibrium price and quantity of dark chocolates to change? Provide a justification for your answer.